

Name: _____

FRACTIONS, DECIMALS AND PERCENTAGES, (FINDING PERCENTAGES OF AMOUNTS)

Date: _____

LO: I can calculate a percentage of a given amount using mental and written methods.

Percentages of amounts

Finding a **percentage of an amount** means finding that **fraction** of the total. 'Of' means multiply.

Method: Divide by 100, then multiply by the percentage. Or find 10% first and build up.

Key methods

●	50% of 80 = 40	half
●	10% of 250 = 25	÷ 10
●	25% of 60 = 15	÷ 4
●	10% of 80 = 8.0 (move 2 places)	only move 1 place
●	30% of 50 = 30 × 50 = 1500	must divide by 100 too

Key idea

To find x% of an amount: divide by 100, multiply by x. Or use 10% building blocks.

$$\text{x\% of A} = \text{A} \times \text{x} \div 100$$

Percentage means per hundred

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - simple percentage

Find 20% of £150.

$$\begin{aligned} 10\% \text{ of } 150 &= 15 \\ 20\% &= 2 \times 10\% = 2 \times 15 \\ &= \text{£}30 \end{aligned}$$

$$= \text{£}30$$

Find 10% first, then double it.

Example 2 - using division

Find 35% of 200.

$$\begin{aligned} 200 \div 100 &= 2 \text{ (this is 1\%)} \\ 2 \times 35 &= 70 \\ 35\% \text{ of } 200 &= 70 \end{aligned}$$

$$= 70$$

Divide by 100 to get 1%, then multiply.

Example 3 - non-calculator

Find 15% of £80.

$$\begin{aligned} 10\% \text{ of } 80 &= 8 \\ 5\% &= \text{half of } 10\% = 4 \\ 15\% &= 8 + 4 = \text{£}12 \end{aligned}$$

$$= \text{£}12$$

Build up from 10% and 5%.

Example 4 - applied

A coat costs £120. There is a 30% discount. How much is the discount?

$$\begin{aligned} \text{Find } 30\% \text{ of } 120 \\ 10\% &= 12, \text{ so } 30\% = 3 \times 12 = 36 \\ \text{Discount} &= \text{£}36 \end{aligned}$$

$$= \text{£}36$$

The discount means you save £36; sale price = £84.

YOUR TURN – PRACTICE (deliberate practice)

1. Simple percentages

Find each percentage.

- a) 50% of 60
- b) 25% of 80
- c) 10% of 350
- d) 75% of 40

2. Using 10% building blocks

Find each percentage of the amount.

- a) 20% of 90
- b) 30% of 150
- c) 40% of 200
- d) 60% of 250

3. Trickier percentages

Use 10% and 5% building blocks.

- a) 15% of 120
- b) 35% of 200
- c) 45% of 80
- d) 5% of 360

4. Word problems

Calculate each amount.

- a) 20% discount on £55
- b) 15% tip on a £40 meal
- c) VAT is 20% on £250
- d) 65% of 120 students passed

5. Mixed

Find each percentage.

- a) 12% of 300
- b) 85% of 60
- c) 2.5% of 400
- d) 17.5% of 200
- e) $33\frac{1}{3}\%$ of 90

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 10% means divide by 10 ☐

Correct answer: _____

Reason: _____

b) 50% of 70 = 140 ☐

Correct answer: _____

Reason: _____

c) 25% is the same as $\frac{1}{4}$ ☐

Correct answer: _____

Reason: _____

d) To find 1%, divide by 10 ☐

Correct answer: _____

Reason: _____

e) $15\% = 10\% + 5\%$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A TV costs **£480**. Shop A offers **25% off**. Shop B offers **£100 off**. Which is cheaper? By how much?

Expression: _____

Simplified: _____

b) A school has **1200 students**. **55%** are girls. **30%** of the girls and **40%** of the boys take the bus. How many students take the bus in total?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

